

Software Requirements Specification

5.1 Introduction

The DVD Lending System is a web application for the CS department media center designed to replace a manual paper log. The primary goal is to provide real-time inventory visibility, track checkout history, and manage borrowing rules.

5.2 Function Requirements

5.2.1 User Authentication

- The system shall authenticate users such as borrowers, staff, and system administrators through an existing campus login system.

5.2.2 Inventory Search and Browse

- Users shall be able to search for DVDs by title.
- Users shall be able to browse the catalog by genre or availability status.

5.2.3 DVD Checkout/Borrowing

- The system shall allow a borrower to check out a DVD if it's currently marked as available.
- The system shall automatically assign a due date for every borrowed DVD.
- The system must prevent other users from borrowing any DVD that is already checked out.

5.2.4 DVD Return

- The system shall allow borrowers to return DVDs and update the status back to "available."

5.2.5 Catalog Management

- Staff shall have the ability to add new DVDs to the catalog.
- Staff shall have the ability to remove existing DVDs from the catalog.

5.2.6 Overdue Monitoring

- Staff shall be able to view a list of all DVDs that are past their due date.

5.2.7 System Configuration

- System administrators shall have access to oversee system configuration and generate general usage reports.

5.3 Performance Requirements

5.3.1 Real-time Status Updates

- The system shall update the availability status of a DVD immediately upon checkout or return to ensure accuracy in the availability data.

5.3.2 Concurrent User Access

- The system shall support multiple simultaneous users browsing and searching the catalog without significant latency.

5.3.3 Authentication Response Time

- The handshake with the campus login system should be processed in under 2 seconds to ensure usability.

5.4 Environment Requirements

5.4.1 Development Environment Requirements

- **Version Control:** A Git-based repository for iterative Agile development.
- **Framework:** A standard web development stack (Next.js) suitable for a simple web application.
- **Database:** A relational database to track DVD status, checkout history, and user roles (Supabase).

5.4.2 Execution Environment Requirements

- **Platform:** The system must be hosted on a web server accessible via standard campus network protocols (Vercel).
- **Client Access:** The application must be compatible with modern web browsers used by students and faculty.
- **Integration:** The environment must have secure network access to the existing campus authentication API.